

Internships

Mathematics of Machine Learning



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Keywords. Sampling, optimal transport, stochastic differential equations.

Targeted departments. Statistics, Mathematics, Computer Science.

Expected output. Review article or conference submission.

Context. Generative AI and sampling both aim to generate samples from a target probability distribution, but they traditionally assume different forms of access to it. In generative AI, the target distribution is represented by a finite set of samples; in classical sampling, it is specified through an unnormalized density. Consequently, the two fields have largely developed in separate communities, despite their shared objective.

State-of-the-art algorithms from both areas are used to power video-generation in the industry, as well as molecular-configurations in computational chemistry. Perhaps surprisingly, all these algorithms—including diffusion and flow models, annealed MCMC, Sequential Monte Carlo, simulated tempering, and parallel tempering—can be described within a common mathematical framework:

1. Choose a probability path $(p_t)_{t \in [0,1]}$ connecting a simple reference distribution at $t = 0$ to the target distribution at $t = 1$.
2. Define transition rules that move samples between different values of t

The efficiency of the resulting algorithm depends critically on the choice of probability path. This internship will study generative modeling and sampling within this unified framework, with a particular focus on designing paths that lead to provably fast convergence.

Our recent work has begun to connect the geometry of probability paths to algorithmic efficiency. Results for annealed importance sampling [Chehab et al., 2024a] and annealed MCMC [Chehab et al., 2024b], together with related developments [Guo et al., 2025, Habring, 2025, Falk et al., 2026], suggest that paths that are short or smooth in an appropriate geometry can yield optimal or near-optimal algorithms.

Goals. The goal of this project is to advance a research program that directly relates the *geometry* of probability paths to the *speed* of state-of-the-art algorithms for realistic, multimodal target distributions. In particular, we will investigate how the Wasserstein geometry of a probability path affects the convergence of annealed MCMC algorithms. The project will proceed through the following stages:

- **Understand current results.** Study and clearly summarize the existing convergence proofs, while becoming familiar with their main mathematical tools. We will first examine two analyses of annealed MCMC in continuous spaces: one based on isoperimetric inequalities [Chehab et al., 2024b] and the other on optimal transport [Guo et al., 2025]. We will

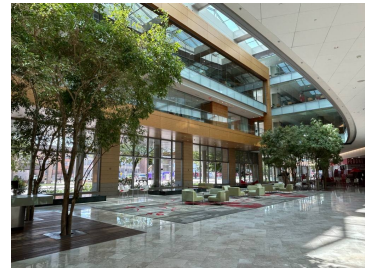
then study the extension of the optimal-transport approach to discrete spaces by [He et al. \[2026\]](#). AI tools are allowed and even encouraged to help assist in understanding existing proofs, but it cannot be used lazily: the candidate will have to show, after the AI has been used, that they master every step of the final proof on their own, without the AI.

- **Extend the analysis to approximate transition kernels.** Build on [He et al. \[2026\]](#) by introducing a term that quantifies the error incurred when the transition kernel must be estimated. Compare the graph geometry in the proof to the graph geometry (pairs of nodes that have Hamming-1 distance have an edge) used in discrete diffusion models.
- **Extend the framework to another algorithm.** Review existing results for the Parallel Tempering algorithm [[Chen et al., 2020](#), [Lee and Shen, 2023](#)]. Investigate whether the same geometric perspective can be used to analyze and design probability paths for the Parallel Tempering algorithm. Relate these findings to the heuristic “round-trip-rate” [[Syed et al., 2021](#)].

Expected outcome. This is a primarily theoretical project with minimal coding. By the end of the internship, the candidate will be familiar with modern mathematical tools for analyzing sampling algorithms in continuous and discrete spaces, including techniques from optimal transport, stochastic processes, and functional inequalities.

At a minimum, the candidate will produce a clear synthesis of current results in the form of a review article or blog post, corresponding to the first stage above. If sufficient progress is made on the extensions proposed in the second and third stages, we will work toward a conference submission.

Environment. The position is within my [research group](#) at the University of Wisconsin–Madison. We work at the intersection of machine learning and applied mathematics, with a particular focus on generative models. The group combines mathematical foundations with practical algorithm development and applications in areas such as brain imaging and molecular science.



Requirements. We are seeking *two* highly motivated candidates, typically a master’s student or advanced undergraduate, with a strong interest in the mathematical foundations of machine learning and sampling. The successful candidate should have:

- a strong background in mathematics or statistics;
- strong familiarity with optimal transport in continuous space
- familiarity with, or interest in, stochastic differential equations, isoperimetric inequalities and spectral gaps, or Markov-chain Monte Carlo;
- strong written English and/or an interest in developing clear scientific writing;

How to apply. Please contact me by email with the object [Internship] Annealing Geometry and Convergence and attach:

1. a CV named `firstname_lastname_CV.pdf`, including a link to your Google Scholar profile, if applicable;
2. one paragraph in `firstname_lastname_cover.pdf`, explaining why you are interested in the position and what you believe you would bring to the project.

References

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